

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (EE) Examination (Backlog)

Nov 2012

EE 307 Electrical Power System – III

Date: 06.11.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70.

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) List out the advantages of per unit system. Taking the generator rating as base, draw [07]
the per unit reactance diagram for the system shown in Figure 1. Reactance of line is 100 ohm.

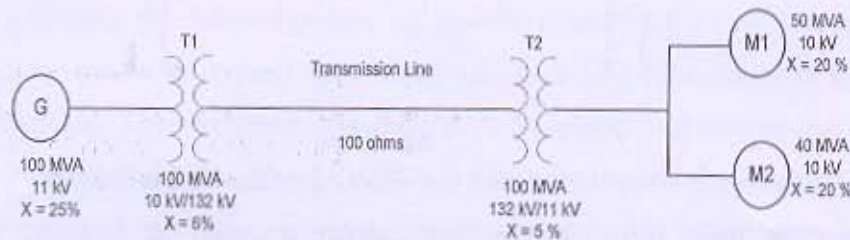


Figure 1

- Q - 2 (a) Derive the equation of the short circuit current for the symmetrical fault using simple [05]
R-L series circuit.
- (b) Figure 2 shows a generating station feeding a 132 kV system. Determine the total [07]
fault current, fault level and fault current supplied by each alternator for a three phase
fault at the receiving end bus. The line is 200 km long.

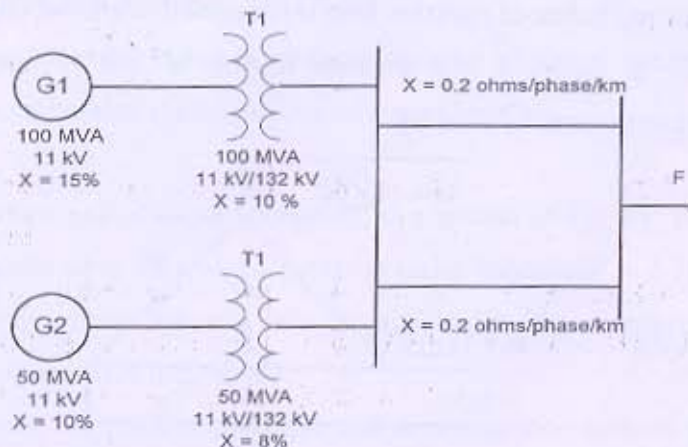


Figure 2

- (c) Define electrical faults and give the classification of faults. Also list out their probability of occurrence. [02]

OR

- Q - 2 (a) Draw and explain the current waveform of the alternator under short circuit condition. [05]
Also explain the sub transient, transient and steady state model of the alternator.
- (b) For the system shown in Figure 3, find (a) fault level for the fault on one of the feeders (near bus) with zero value of reactance X (b) the reactance of the current limiting reactor X to limit the fault level to 800 MVA for a fault on one of the feeders 800 MVA for a fault on one of the feeders. [07]

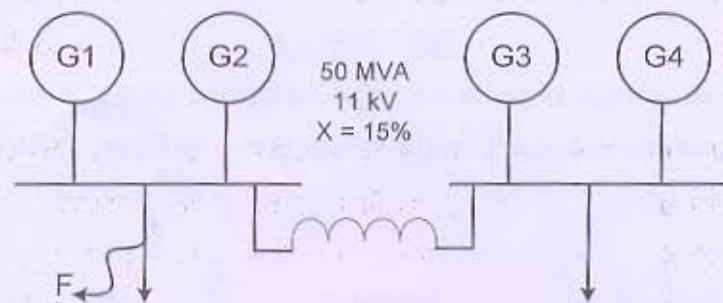


Figure 3

- (c) Explain factors to be considered for the selection of the circuit breaker. [02]
- Q - 3 (a) Derive the three phase voltage and current equation in terms of symmetrical components and prove that sum of powers of three symmetrical components equals the three phase power. [05]
- (b) Derive the equation for the fault current for single line to ground fault. [06]
- (c) Draw the zero sequence networks for all types of three phase transformers. [03]

OR

- Q - 3 (a) Derive the equation of fault current for line to line fault. [04]
- (b) A delta connected balanced resistive load is connected across an unbalanced 3 phase supply shown in Figure 4 with currents in line 'a' and 'b' specified. Find the symmetrical components of currents. [05]

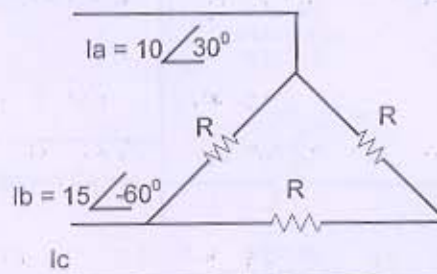


Figure 4